The examples in this chapter are not all covered by the Free Tier. A warning message appears when an example incurs costs. Nevertheless, as long as you don't run all other examples longer than a few days, you won't pay anything for them. Keep in mind that this applies only if you created a fresh AWS account for this book and nothing else is going on in your AWS account. Try to complete the chapter within a few days; you'll clean up your account at the end.

10.1 Starting a MySQL database

In the following section, you will launch the infrastructure required to run WordPress on AWS. In this chapter, we will focus on the MySQL database provided by RDS, but you can easily transfer what you learn to other database engines such as Aurora, PostgreSQL, MariaDB, Oracle, and Microsoft SQL Server, as well as to applications other than WordPress.

When you follow the official "How to Install WordPress" tutorial (see http://mng.bz/G1vV), one of the first steps is setting up a MySQL database. Formerly, you might have installed the database system on the same virtual machine that also runs the web server. However, operating a database system is not trivial. You have to implement a solid backup and recovery strategy, for example. Also, when running a database on a single virtual machine, you are introducing a single point of failure into your system that could cause downtimes of your website.

To overcome these challenges, you will use a fully managed MySQL database provided by RDS. AWS provides backup and restore functionality and offers database systems distributed among two data centers as well as the ability to recover from failure automatically.